Adaptive Task-Oriented Semantic Communication for UAVs With Joint Trajectory and Resource Optimization

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***Abstract*—The transmission of large-volume sensed images during uncrewed aerial vehicle (UAV) inspections poses significant challenges due to the UAV's limited bandwidth and energy constraints. This paper proposes a UAV-adaptive task-oriented semantic communication (UA-TSC) framework for low-altitude inspection tasks, which extracts task-relevant semantic features on the UAV and jointly optimizes flight trajectory and communication resource allocation to enhance transmission efficiency and task performance. We first develop a joint source-channel coding (JSCC) model for UAV semantic communication and formulate a unified optimization problem that maximizes a weighted sum of task execution accuracy, energy consumption, and trajectory reward under system constraints. To efficiently solve this problem, we decompose it into two coupled subproblems. The trajectory design subproblem is tackled using a Diffuser-based offline reinforcement learning algorithm, while the resource allocation subproblem is handled by a proximal policy optimization (PPO)-based algorithm. Simulation results demonstrate that the proposed semantic transmission scheme improves task execution accuracy by 15% over the local inference (LI) scheme and achieves a 5-fold improvement over the raw image offloading (RIO) scheme under low-SNR condition. Furthermore, the PPO-based resource allocation strategy consistently outperforms two baseline methods, while the learned trajectory optimization algorithm yields smoother paths and better adaptability across diverse environments.**

***Index Terms*—Uncrewed aerial vehicle (UAV), semantic communications, computation offloading, energy efficiency, task-oriented, diffusion model, proximal policy optimization (PPO).**

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I. Introduction

With the growing demand for emerging services, the 6G standards are gradually advancing, encompassing a wide range of application scenarios such as immersive communications, ultra-massive connectivity, ultra-reliable and low-latency communications (URLLC), integration of communication, sensing and computing, and ubiquitous connectivity [1] [2]. Among these applications, UAVs have emerged as a key enabler in 6G networks, offering high mobility, flexible deployment, and broad coverage for diverse communication and sensing tasks [3].

As an emerging economic paradigm, the Low-Altitude Economy (LAE) leverages low-altitude airspace (typically below 1000 meters above ground) to enable diverse aerial activities, thereby generating substantial commercial and societal value [4]. These activities, supported by crewed and uncrewed aerial vehicles and integrated with Non-Terrestrial Networks (NTNs), have been widely adopted in various domains such as the Internet of Things (IoT), artificial intelligence (AI), transportation, logistics, tourism, agriculture, and disaster monitoring, due to their advantages in high mobility, low cost, and air-space-ground integrated communications [5] [6] [7] [8]. Within this framework, UAVs serve as a pivotal enabler of LAE, demonstrating significant potential in diverse applications. For instance, UAVs can be deployed for traffic monitoring to assist in accident avoidance and traffic management [9] or for power-line inspection to detect damage in real time [10].

One of the most representative UAV applications is image/video transmission, where UAVs capture visual data and transmit it to a base station (BS) for further analysis to support mission-critical decision-making [11]. However, UAV image transmission faces several challenges. Since UAVs operate without stable power supply in the air, their available energy is strictly limited. Meanwhile, transmitting large volumes of images/videos will lead to considerable communication resource overhead, which limits the efficiency of traditional communication and resource allocation schemes [12].

Semantic communication (SemCom), as a novel communication paradigm, has been recognized as a key enabling technology for 6G networks [13] [14] [15]. By extracting semantic features from source data, it achieves efficient information transmission. For end-to-end semantic communication systems, a variety of

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deep learning-based approaches have already been studied in the academic community. For text sources, studies have proposed JSCC schemes based on the Transformer architecture [16] [17], as well as semantic encoding and decoding schemes built upon bidirectional long short-term memory (BiLSTM) networks [18]. For image sources, researchers have developed JSCC methods using models such as ResNet [19] and Transformer [20] [21] to effectively extract and transmit semantic features. These advances demonstrate the potential of semantic communication to bridge communication and intelligence, paving the way for task-oriented transmission with reduced communication overhead in future wireless networks.

In most practical scenarios, users do not require complete image data but only the task-relevant information to accomplish specific objectives. Task-oriented semantic communication focuses on transmitting only the information that is essential for accomplishing specific downstream tasks, rather than reconstructing the original data at the receiver [22] [23] [24]. This paradigm has been widely applied in various intelligent perception and inference tasks, such as image classification [25], image retrieval [26], speech recognition [27], visual question answering (VQA) [28]. UAVs which frequently operate under dynamic channels and strict energy constraints, are also characterized by unique task-specific performance requirements. Therefore, it is essential to investigate task-oriented semantic communication for UAV systems [29].

Resource allocation plays a vital role in ensuring communication efficiency, reliability, and cost-effectiveness. Unlike traditional communications that rely on quality of service (QoS) indicators such as data rate and throughput, task-oriented semantic communication emphasizes task-related quality of experience (QoE) metrics as optimization objectives [30]. By transmitting only task-relevant information, SemCom outperforms traditional methods and adapts to heterogeneous user and mission demands [31][32] [33]. Moreover, deploying semantic models on devices requires balancing computational and communication resources [34], making efficient resource allocation strategies essential to realize the full potential of SemCom in dynamic environments.

With the development of reinforcement learning (RL), it has been increasingly applied to resource allocation in dynamic communication scenarios [35] [36], such as deep deterministic policy gradient (DDPG) [32], proximal policy optimization (PPO) [37], multi-agent proximal policy optimization (MAPPO) [38], soft actor-critic (SAC) [39], double deep Q-networks (DDQN) [40]. Beyond RL-based resource optimization, recent studies have started to explore more advanced AI paradigms, including generative AI [41] [42] [43] [44]. In particular, generative AI-assisted secure communications in integrated space-air-ground networks have recently emerged as an active research direction [45]. Moreover, interactive artificial intelligence (IAI) has been incorporated into communication networks to enable intelligent network management and optimization [46].

In UAV aerial imaging, the dynamic mobility and limited energy of UAVs destabilize image transmission and reduce task accuracy, which further necessitates holistic efficiency optimization to extend flight duration, posing challenges for semantic communication resource allocation [47]. However, existing UAV semantic communication studies mainly optimize semantic-level metrics, while paying limited attention to the task-oriented requirements of UAV applications [48]. Moreover, current studies mainly focus on communication parameters or the number of transmitted semantic feature symbols, whereas the lightweight and adaptive structural optimization of semantic models remains largely unexplored.

To address these challenges, this paper proposes a UAV-adaptive task-oriented semantic communication (UA-TSC) framework. Specifically, a joint source-channel semantic coding model is designed by incorporating the dynamic characteristics of UAV communication channels, while deep reinforcement learning (DRL) is employed to optimize resource allocation strategies, thereby balancing task execution accuracy with communication-computation resource consumption. Furthermore, inspired by the Diffuser-based trajectory optimization approach [49] [50], we adopt an offline DRL algorithm to generate smooth UAV trajectories that comply with physical dynamics, ultimately enhancing task performance and overall system efficiency.

The contributions of this paper are summarized as follows.

- We propose a UAV-adaptive task-oriented semantic communication framework tailored for image transmission and tasks. The framework adopts a Transformer-based semantic encoder at the UAV to extract task-relevant semantic features and incorporates a model lightweight design to reduce on-board computational complexity and energy consumption.
- We design a joint source-channel semantic coding network that characterizes the relationship among task execution accuracy, semantic model depth, and the number of transmitted semantic feature symbols. Based on this model, we formulate a unified optimization problem that jointly designs the UAV trajectory and allocates computation and communication resources to strike a favorable tradeoff between energy consumption and task execution accuracy, while improving network resource utilization.
- We decompose the original problem into two coupled subproblems and develop tailored learning-based solvers. A PPO-based algorithm is designed for joint resource allocation to balance classification accuracy and energy consumption, whereas a Diffuser-based trajectory optimization scheme plans UAV paths that trade off proximity to the BS and mission completion under waypoint constraints.
- Extensive simulations show that, under low-SNR conditions, the proposed UA-TSC scheme improves task execution accuracy by 5-fold and 15% over raw image offloading and local inference baselines, respectively. Moreover, the PPO-based resource allocation algorithm consistently outperforms SAC and TD3 algorithm, while the Diffuser-based trajectory optimizer yields smoother

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trajectories, higher task performance, and stronger generalization capability in unseen environments.

Our paper is organized as follows. In Section II, we provide an overview of related research on task-oriented semantic communication systems and resource allocation for UAV semantic communication. In Section III, we present our system model and describe the problem we are addressing. In Section IV, we propose a trajectory optimization algorithm based on Diffuser and a resource allocation algorithm based on PPO to solve the problem. Section V contains our numerical results, and we conclude this paper with Section VI.

II. Related Work

A. Task-Oriented Semantic Communication Systems

1) Neural Semantic Communication Architectures: In recent years, with the continuous development of deep learning technology, research on semantic communication systems has increasingly become a focus of attention. At present, numerous deep learning-based semantic communication methods have been proposed, for text [16] [51], image [20] [21] [52] [53] [54] [55] [56] [57] [58] [59] and video transmission [60].

For text transmission, [16] propose a deep learning based semantic communication system named DeepSC. Due to limited computing capability of IoT devices, [51] propose a lite distributed semantic communication system based on DL, named L-DeepSC, for text transmission with low complexity. The L-DeepSC becomes affordable for IoT devices and reduces the bandwidth required for model weight transmission between IoT devices and the cloud/edge.

For image or video semantic transmission, early studies mainly focused on deep learning-based JSCC frameworks. [53] proposed Deep JSCC, where CNN-based encoders and decoders are jointly trained end-to-end to minimize image reconstruction MSE, effectively alleviating the cliff effect of conventional separation-based systems. Building upon this idea, [54] introduced DeepJSCC-f, which incorporates channel output feedback to improve reconstruction quality under fixed-length transmission and reduce latency for variable-length transmission. They further extended these designs to DeepJSCC-l [55], a family of CNN-based JSCC schemes that support multi-channel transmission and flexible bandwidth adaptation. [60] proposed DeepWiVe, which is the first end-to-end JSCC video transmission scheme that leverages RL to optimize the allocation of limited available channel bandwidth across video frames for maximizing overall visual quality.

Beyond JSCC, recent studies have increasingly focused on semantic-level representations and transformer-based architectures. [56] proposed an Intelligent Semantic Source Coding (ISSC) framework for large-scale visual data transmission in Internet of Vehicles (IoV) scenarios, leveraging a Swin Transformer-based multiscale semantic feature extractor to enhance semantic expressiveness. Along similar lines, Swin Transformer-based semantic communication schemes that explicitly account for wireless channel effects were shown to outperform CNN-based Deep JSCC methods [57]. Robust semantic communication under semantic noise was investigated in [20] via adversarial training and masked autoencoders, while [21] introduced a nonlinear transform source-channel coding (NTSCC) approach that integrates nonlinear transforms with deep JSCC to improve end-to-end performance. Moreover, knowledge-enhanced semantic communication frameworks were developed in [52] to facilitate semantic reasoning at the receiver. In addition, [58] proposed an RL-driven adaptive semantic coding scheme with semantic bit allocation and generative decoding, enabling bandwidth-sensitive and semantically rich image transmission. Against physical-layer adversarial attacks, [59] proposes a physical-layer adversarial perturbation generator SemAdv, along with an adversarial training strategy SemMixed to enhance the robustness of semantic communications.

2) Task-Oriented Semantic Communication: In some scenarios, the purpose of semantic communication is no longer to recover and reconstruct the original information, but rather to enable the execution of downstream intelligent tasks at the receiver. Existing studies have investigated task-oriented semantic communication frameworks [19] [26] [28] [61] [62] [63] [64] [65] [66].

Some works have investigated single-modality tasks, such as question answering, image classification, and semantic segmentation. For sentiment analysis and question answering tasks, [61] propose a semantic extraction mechanism that converts transmitted texts into semantic triplets, which are further compressed through a tailored semantic filtering strategy. For image data, [19] presented a deep learning-based JSCC scheme for IoT devices to wirelessly transmit data to the BS for subsequent recognition tasks. Targeting surveillance scenarios with wireless cameras or UAVs, [26] proposed a compression scheme for DNNs used in image retrieval tasks and adopted JSCC. Compared with traditional coding schemes, their approach improves both accuracy and coding speed. [62] proposes a learning-based communication scheme that leverages the information bottleneck (IB) framework to formalize the rate-distortion tradeoff between the informativeness of encoded features and inference performance. [63] proposes a novel adaptive semantic reconstruction-based SC (SCASR) scheme to improve semantic segmentation performance and adaptability via a semantic importance-based compression mechanism and an attention-integrated adaptive semantic network. [64] proposes a multi-task semantic communication framework for connected and autonomous vehicles (CAVs), along with a convolutional auto-encoder (CAE) for semantic encoding of road traffic signs, and a task-oriented semantic decoder dedicated to image reconstruction and classification tasks.

Beyond single-modal settings, multimodal semantic communication has also attracted considerable attention. A multimodal multi-user semantic communication scenario for VQA was considered, where some users send information in the form of images, while others ask questions about these images in text form [28]. Since multimodal information among users is correlated, the authors proposed a semantic communication system MU-DeepSC, which leverages deep neural networks to perform VQA tasks. Furthermore, [65] introduced a multimodal

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multi-user semantic communication system based on a unified Transformer framework. The authors developed the DeepSC-IR, DeepSC-MT, and DeepSC-VQA models for image retrieval, machine translation, and VQA tasks, respectively. [66] proposed a distributed task-oriented communication network (DTCN), which is based on the latest advances in multimodal semantic transmission and edge intelligence. These studies demonstrate that multimodal semantic communication effectively exploits cross-modal correlations among users to improve task-oriented transmission efficiency and inference performance.

3) Resource-Aware and Energy-Efficient Semantic Communication: Different from conventional communication systems, resource allocation in semantic communications requires explicit consideration of semantic relevance and task objectives. Recent studies have proposed resource allocation frameworks for semantic communications, with particular emphasis on different data modalities and tasks [30] [31] [67] [68] [68] [69] [70] [71] [72] [73].

For text semantic communication framework DeepSC, [30] formulates a semantic-aware resource allocation problem with the objective of maximizing the overall semantic spectral efficiency of all users, and derives its optimal solution, whose effectiveness and feasibility are further validated. Furthermore, [31] investigates semantic-aware resource allocation in a multi-cell multi-task network by formulating a QoE maximization problem subject to constraints on the number of transmitted semantic symbols, channel assignment, and power allocation, for which a matching-theory-based solution is developed.

Several studies introduce optimization framework for knowledge-assisted semantic communication. Specifically, [67] proposes a performance optimization framework for joint optimization of knowledge graph (KG)-modeled partial semantic information and resource block (RB) allocation, maximizing a composite semantic similarity metric under wireless resource, delay and semantic communication performance constraints. In iSemCom-enabled heterogeneous networks, [68] introduces a semantic metric termed STM and investigates the joint user association (UA) and bandwidth allocation (BA) problem by maximizing STM under knowledge base (KB) matching and wireless bandwidth constraints, leading to a two-stage solution. Building on [68], [69] further defines two general SemComNet scenarios, namely perfect knowledge matching-based SemComNet and imperfect knowledge matching-based SemComNet.

To accommodate different scenarios and task requirements, a variety of resource allocation frameworks for task-oriented and multi-modal semantic communications have been proposed. [70] studies semantic-aware resource allocation in an EH-CR-NOMA scenario by optimizing QoE through a trade-off between semantic rate and semantic fidelity. [71] propose a video semantics-based resource allocation algorithm for spectrum multiplexing scenarios, targeting both vehicle-to-infrastructure (V2I) semantic understanding and vehicle-to-vehicle (V2V) information transmission. [72] design a collaborative multi-server framework in which semantic content selection and resource block allocation are jointly optimized to minimize average transmission latency under image-to-graph semantic similarity constraints, solved via a value-decomposition-based entropy-maximized multi-agent deep reinforcement learning approach. [73] proposes an importance-aware JSCC framework with a JSCT mechanism that saves resources by selectively transmitting task-important features, and defines a novel TOSSE metric while developing a K-PPO algorithm to optimize importance-aware resource allocation for maximizing total TOSSE.

B. Resource Allocation and Trajectory Optimization for UAV Communication

1) Model-Based Joint Trajectory and Resource Optimization: Existing UAV communication systems primarily focus on MEC scenarios. In these systems, model-based optimization methods are widely adopted to transform inherently non-convex optimization problems into a series of tractable sub-problems, which are then solved using various mathematical techniques. According to their main optimization objectives and system design considerations, the existing works can be broadly categorized into four groups: energy-oriented optimization [74] [75] [76] [77], latency-driven task execution [78] [79] [80], joint energy-performance tradeoff optimization [81] [82], and advanced technology-assisted UAV-MEC architectures [83] [84] [85] [86].

Owing to the limited onboard energy of UAVs, some existing studies primarily focus on minimizing the energy consumption of UAV-assisted communication and computing systems. [74] investigates a UAV-enabled MEC system based on ground BSs. Under the constraints of UAV mobility and ground BS energy budgets, it makes per-slot decisions by jointly optimizing resource allocation, UAV trajectory planning, and task assignment to minimize energy consumption, where a successive convex approximation (SCA) technique is applied to obtain suboptimal solutions. [75] considers a scenario where a single UAV serves multiple users and jointly optimized the UAV's flight trajectory, task allocation, and CPU frequency to minimize energy consumption of the overall system. The problem is solved using a discrete linear state-space approximation and the concave-convex procedure (CCCP) algorithm. [76] minimizes UAV energy consumption by jointly optimizing flight trajectory and task offloading and decomposes the original problem into two sub-problems: interval partitioning and trajectory planning, which are modeled as a semi-discrete optimal transport problem and a traveling salesman problem for solution respectively. [77] formulates an optimization problem to minimize the total energy required by the UAV by jointly optimizing CPU frequency, offloading amount, transmission power, and UAV trajectory. It designs an algorithm based on SCA and an algorithm based on Decomposition and Iteration (DAI).

For delay-sensitive services, it is also necessary to explicitly optimize system latency as well as task execution performance. In [78], a UAV-ground cellular edge system is studied, where UAV flight trajectories and task offloading schedules are jointly optimized under the UAV's maximum speed and the BS's computational capacity constraints to minimize task completion latency. [79] jointly optimize offloading time, flight trajectory,

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and resource allocation variables to minimize the total task completion time; and propose an iterative algorithm based on block coordinate descent BCD and SCA. Regarding the research where UAVs act as relay nodes, [80] jointly optimize user transmission power control, task scheduling, and UAV trajectory to minimize the total user delay; and adopt BCD technology for solution.

Moreover, several works investigate the trade-off between energy consumption and task performance, aiming to achieve balanced system optimization under resource-constrained UAV operations. Focusing on autonomous detection and tracking in maritime environments, [81] jointly optimizes image resolution, transmit power, offloading ratios, and CPU frequency to maximize a weighted sum of energy efficiency and recognition accuracy. [82] further consider inter-task dependencies, jointly optimize task offloading, resource allocation, and flight trajectory; and propose an alternating iterative method based on BCD.

Recent studies extend UAV MEC optimization to more advanced system architectures, such as IRS-assisted networks, relay-enabled UAV systems, and wireless power transfer-aided scenarios and so on. [83] investigates the scenario of a single UAV and multiple ground base stations, decomposes the joint optimization problem into trajectory planning and task offloading subproblems, adopting the differential evolution algorithm to plan the UAV path, and the Earliest Deadline First (EDF) algorithm to handle task offloading. [84] considers an RIS-assisted UAV network. The authors jointly optimize the UAV trajectory, task offloading, caching decisions, and RIS phase shifts, and proposed a two-layer iterative algorithm based on the Dinkelbach method and the BCD technique. [85] further focus on the single-UAV multi-user scenario combined with IRS, jointly optimize bit allocation, transmission power control, IRS phase shifts, and trajectory planning, and design an alternating optimization algorithm based on Dinkelbach and BCD technologies. [86] consider the WPT-aided single-UAV relay scenario, jointly optimize trajectory planning, communication and computing resource allocation, and time scheduling decisions to maximize the minimum task computing data volume of all active devices, and obtain a suboptimal solution through a two-layer cyclic iterative algorithm based on BCD and SCA.

2) Learning-Based Joint Trajectory and Resource Control: Under dynamically varying channel conditions, learning-based optimization strategies have emerged as an effective solution for UAV networks, owing to their strong adaptability and decision-making capability. Existing studies can be broadly categorized according to the adopted learning paradigm into single-agent reinforcement learning [89] [90] [91] [92] [98] [99], multi-agent reinforcement learning [100] [101] [102] [103] [104] [105] [106] [107] [108], and offline reinforcement learning [109] [110].

Recent studies have applied RL algorithms to UAV task offloading and trajectory optimization. [89] developed an RL-based UAV-MEC offloading framework that selects optimal MEC servers and UAV-task matching strategies to minimize energy consumption and processing delay, using a Q-learning algorithm. [90] jointly optimized UAV trajectory and computation offloading via a DDPG framework enhanced with a population-diversity particle swarm optimization (PSO) method to reduce energy-delay cost. [91] proposed an independent PPO (IPPO) framework to jointly optimize task offloading ratios and UAV trajectories while achieving queue stability using Lyapunov drift-plus-penalty methods. [92] jointly optimized UAV trajectory, task offloading decisions, and time allocation ratios under queue stability constraints to maximize the time-averaged secrecy capacity, and solved the problem using a DQN framework. [98] proposed a PPO-based task offloading optimization strategy to provide computing offloading services for terminal devices in emergency scenarios such as natural disasters. The proposed algorithm achieves significantly better convergence speed and stability than conventional methods and effectively reduces latency in dynamic environments. [99] further extended this work by developing a joint optimization framework for dynamic environments. Their PPO-based approach optimizes multiple UAV control variables, including flight angle and speed, in a discretized action space, exhibiting superior convergence when dealing with non-convex problems and high-dimensional state spaces.

For multi-UAV scenarios, some studies adopt multi-agent reinforcement learning (MARL) to enable cooperative decision-making for trajectory planning, resource allocation, and collision avoidance. [100] proposed a reinforcement learning-based sensing-and-transmission framework for multi-UAV and base station systems, where the system is modeled as an MDP with a three-stage protocol consisting of beaconing, sensing, and data transmission to accelerate learning. [101] developed a DQN-based scheme for cooperative multi-UAV communications to maximize the system sum rate under a minimum user coverage constraint, using a dueling DQN architecture. [102] introduced a deep reinforcement transfer learning method to enable knowledge sharing among UAVs, significantly improving learning efficiency and performance. [103] proposed a MADDPG-based joint target assignment and path planning framework that finds collision-free shortest paths for multiple UAVs. [104] studied 3D deployment optimization in multi-UAV networks using deep reinforcement learning to maximize QoE, defined as a weighted sum of data rate and delay. [105], [106], [107] applied reinforcement learning techniques to UAV trajectory design and optimization.

For the high-dimensional state-action space and multi-objective optimization problems in UAV networks, combining offline reinforcement learning or conservative reinforcement learning can improve learning efficiency, optimization performance, and system stability. [109] target the UAV-aided wirelessly powered MEC network, jointly optimize UAV 3D deployment, beamforming, and resource allocation to maximize system throughput, and propose a hybrid optimization framework integrating methods such as polyhedral annexation, multi-objective evolution, DDPG algorithm, and Lagrange duality. [110] focus on the scenario where multiple UAVs collaboratively collect data from ground IoT devices, jointly optimize UAV flight trajectories and device scheduling strategies to minimize the weighted sum of AoI and transmission power; and design a solution framework integrating Conservative Q-Learning (CQL) and model-free reinforcement learning.

Overall, these studies on UAV communications mainly focus on MEC networks, where computation offloading constitutes the

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core research problem. However, these approaches essentially rely on conventional communication paradigms, which often lead to substantial communication resource wastage due to the transmission of large volumes of raw image or video data. Moreover, under highly dynamic channel conditions, the communication quality can suffer significant degradation. Therefore, it is necessary and important to explore semantic communication paradigms for UAV networks.

C. Semantic Communication in UAV Networks

1) Task-Oriented UAV Semantic Communication Framework: Considering the highly dynamic channel conditions and the communication and computation constraints of UAVs, recent studies have developed end-to-end semantic communication frameworks tailored for UAV applications [11] [12] [95] [111] [112].

For scene classification tasks, [11] proposed a new aerial image transmission paradigm that divides images into semantic blocks. A semantic model deployed on the UAV selects blocks for transmission according to the instantaneous channel state, and DRL is employed to balance transmission latency and classification accuracy. Based on triplet scene graphs, [95] designed an energy-efficient task-oriented UAV semantic communication framework and further proposed a novel personalized semantic encoder to meet users' demands according to their individual interests. [111] proposed a KB-driven task-oriented image semantic communication scheme and introduced an improved weighted LPIPS metric to achieve high compression ratio, high-quality image reconstruction, and effective bandwidth limitation resolution. [112] leverages TinyML to perform semantic mapping for heterogeneous multimodal communication and UAV task execution processes, thereby improving the reliability and efficiency of multimodal communication for UAV tasks. [12] proposed a UAV cognitive semantic communication system by exploiting KG. Moreover, a multi-scale compression network is designed for semantic compression to reduce data transmission volume while guaranteeing the detection performance.

2) Optimization of UAV Semantic Communication: Existing UAV semantic communication systems, tailored to different application scenarios, mainly focus on optimizing flight safety and control reliability [40] [113] [114], semantic QoE [115] [47] [48] [116] [117], latency and energy consumption [96] [97] [118], and task-level and system-level performance [119] [120] [121] [122].

To enhance UAV safety and flight control, [40] proposed a task-oriented semantics-aware (TOSA) communication framework for C&C transmission and define its value of information based on both the similarity and age of information (AoI) of C&C signals. The authors also proposed a DRL algorithm to maximize the TOSA information. [113] proposes a goal-oriented semantic communication robot control (GSRC) method. At the transmitter, a deep reinforcement learning algorithm is presented to generate optimal C&C data. At the receiver, a queue scheduling mechanism based on value of information (VoI) and AoI is designed. [114] proposes an innovative multi-agent deep reinforcement learning (MADRL) framework called Graph Attention Exchange Network (GAXNet) for the real-time control of multiple uncrewed aerial vehicles (UAVs) and collision avoidance between UAVs.

Some studies have defined semantic transmission quality or QoE as the primary optimization objective. [115] developed a multi-modal semantic communication framework for UAV image-sensing task-oriented scenario. Furthermore, they formulated a semantic-aware resource allocation problem based on novel QoE and proposed a hybrid decision-controlled multi-UAV DRL scheme to solve it. In [47], semantic resource allocation is formulated based on approximate semantic entropy and semantic rate, aiming to minimize the average cost. In [48], the authors jointly optimize subchannel allocation, semantic symbol assignment, UAV beamforming, and trajectory to maximize semantic spectrum efficiency (SSE) through a DDQN-DDPG-based solution. [116] investigates a UAV semantic communication system equipped with multiple intelligent reflecting surfaces (IRSs), and improves the secure semantic transmission rate (SSR) by optimizing UAV trajectories, IRS reflection coefficients, the number of semantic symbols, and transmission power. [117] introduces a similarity metric for image reconstruction and jointly optimizes trajectory and semantic information selection using a actor-critic with entropy maximization (SAC-EM) algorithm.

Some works focus on reducing system latency or energy consumption in semantic communication system. [96] studies the UAV-assisted digital twin synchronization problem, models it as an average latency minimization task, and solves it via Lyapunov optimization combined with DRL. A hybrid decision-control DRL algorithm (HDCD-SC) is proposed to solve the problem. [97] developed a multi-UAV semantic communication framework to jointly optimize UAV trajectories, channel allocation, and semantic information selection, where a value decomposition-based multi-agent deep reinforcement learning (VD-MADRL)-based solution effectively lowers transmission delay and energy consumption under practical constraints. [118] proposes an energy-efficient uploading mechanism in the UAV-assisted semantic awareness data collection system and minimizes the maximum energy consumption of IoT devices via jointly optimizing user scheduling and UAV trajectory.

With the deeper integration of semantic communication with edge computing and complex task execution, recent studies have further adopted task completion efficiency or joint system performance improvement as the optimization objectives. [119] proposes a task-oriented semantic communication framework, namely the visual question answering-based maritime semantic communication system (VQA-MSCS), aiming to optimize operational efficiency. [120] proposes integrating semantic communication with UAVs to enhance mobile edge computing (MEC) against jamming attacks, and leverages DRL to adjust semantic task and resource scheduling strategies, aiming to reduce task completion time and improve SSE. [121] utilizes UAV-aided semantic communication for data collection of metaverse users, and proposes a hybrid-action RL framework to make decisions on semantic model scale, channel allocation, transmit power and UAV trajectory, thereby improving transmission efficiency. [122] introduced semantic communication paradigm is into UAV

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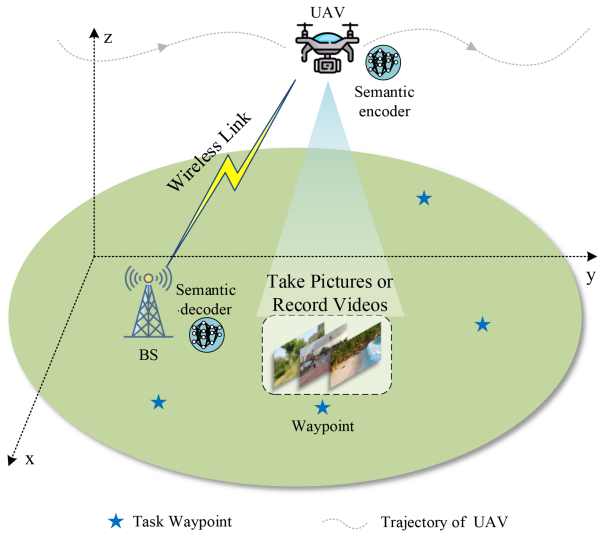


Fig. 1. UAV-assisted image transmission scenario.

swarms, a hierarchical federated learning (HFL) framework is adopted for semantic model training, and a novel cluster management mechanism is designed to address the dynamic dropout problem.

Overall, most existing works focus their optimization objectives on semantic entropy, semantic similarity, or semantic spectrum efficiency, with relatively little attention paid to the personalized requirements of UAV task execution scenarios. In energy-constrained UAV environments, enhancing task-oriented perception and inference accuracy is more aligned with practical needs. In addition, limited effort has been devoted to the lightweight design of semantic model architectures themselves, with most works focusing only on optimizing the number of semantic feature symbols or communication parameters. In this paper, we explore the structural optimization of semantic models, specifically through adaptive selection of the depth of the semantic model. This enables a balance between task performance and energy consumption, providing a more comprehensive and effective optimization perspective for semantic communication.

III. System Model

A. Semantic Communication System for UAV-Based Image Acquisition

This paper considers a scenario where an UAV conducts image acquisition over a designated area and transmits the collected data to a ground BS for processing. The UAV-assisted image transmission scenario is illustrated in Fig. 1, comprising a UAV U and a ground BS G. The UAV flies at a low-altitude height H, and departs from a starting point, passes through a set of intermediate waypoints, and ultimately arrives at the destination point. The entire mission execution is divided into T discrete time slots, during each of which the UAV's position and velocity are assumed to remain constant. The ground BS is assumed to remain stationary throughout the UAV's mission.

During each time slot, the UAV captures images or video and processes them locally using a semantic feature extraction model deployed on the UAV, thereby compressing the volume of data to be transmitted. Semantic models of varying depth and task execution accuracy are deployed on the UAV, forming a set of selectable semantic models denoted as $\mathcal{M}$, and let $\mathcal{N}$ denote the set of possible numbers of semantic feature symbols to be transmitted. Based on the current channel conditions and available resources, the UAV selects an appropriate semantic model, determines the number of semantic feature symbols to transmit, and allocates transmission power accordingly. The selected semantic feature symbols are then transmitted to the BS, where a semantic decoder is executed to infer the task based on the received symbols, enabling functions such as image classification and recognition, and ultimately yielding the final output results.

The system framework of task-oriented semantic communication for UAVs is illustrated in Fig. 2. At the transmitter side (UAV), the image preprocessing module first processes the captured images into a unified dimension, producing image information $\mathbf{s}$ as the input to the semantic encoder. The semantic encoder is designed based on the vision Transformer model DeiT [123], consisting of multiple Transformer encoder layers, which extract semantic features from the images. The channel encoder comprises multiple fully connected layers and is responsible for generating semantic feature symbols to be transmitted over the channel. The process of image information $\mathbf{s}$ passing through the semantic encoder and the channel encoder can be expressed as

$$\mathbf{x} = C_{\boldsymbol{\beta}}(S_{\boldsymbol{\alpha}}(\mathbf{s}))\,, \tag{1}$$

where $S_{\boldsymbol{\alpha}}(\cdot)$ and $C_{\boldsymbol{\beta}}(\cdot)$ denote the semantic encoder and channel encoder functions, respectively, with their corresponding neural network parameters $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$. The number of semantic feature symbols generated by the channel encoder for transmission is K, thus $\mathbf{x} \in \mathbb{C}^{1\times K}$.

The semantic feature symbols $\mathbf{x}$ are transmitted through a wireless channel, and the ground BS receives the signal

$$\mathbf{y} = h(t)\mathbf{x} + \mathbf{n}\,, \tag{2}$$

where $h(t)$ represents channel coefficient between the BS and UAV and $\mathbf{n}$ represents additive white Gaussian noise (AWGN).

At the receiver side (BS), upon receiving $\mathbf{y}$, the channel decoder composed of multiple fully connected layers recovers the semantic information $\hat{\mathbf{z}} = C_{\boldsymbol{\gamma}}^{-1}(\mathbf{y})$ where $C_{\boldsymbol{\gamma}}^{-1}(\cdot)$ denotes the channel decoder function with neural network parameters $\boldsymbol{\gamma}$. Then, the obtained semantic information $\hat{\mathbf{z}}$ is processed by the semantic decoder, which consists of a fully connected layer for the recognition task inference, directly producing the final classification output $P = S_{\boldsymbol{\phi}}^{-1}(\hat{\mathbf{z}})$, where $S_{\boldsymbol{\phi}}^{-1}(\cdot)$ denotes the semantic decoder function and $\boldsymbol{\phi}$ is neural network parameters.

B. Channel Model

Since UAVs usually operate at relatively high altitudes, the air-to-ground link is typically dominated by the line-of-sight (LoS) component, while the effects of shadowing and small-scale fading are comparatively less significant. Therefore, the channel between the UAV and the BS is modeled as a LoS channel The

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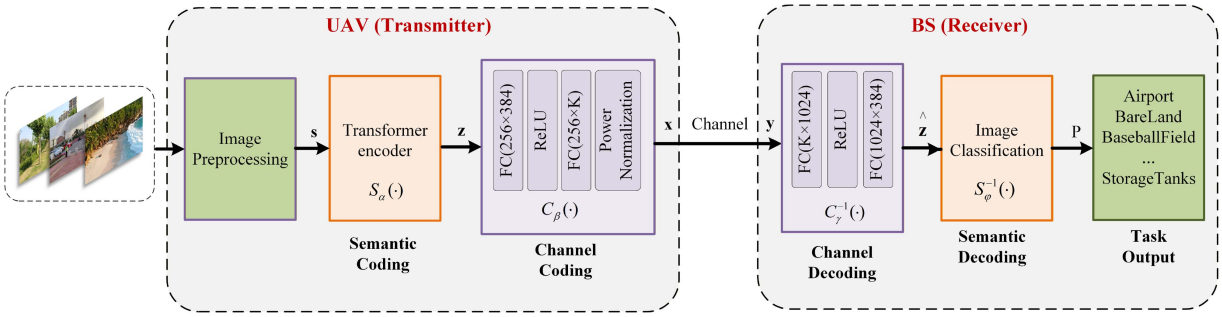


Fig. 2. The framework of UAV task-oriented semantic communication system.

channel coefficient $h(t)$ is related to the distance $d(t)$ between the UAV and the BS. Let $w_U(t) = [x(t), y(t), H]^T$ denote the 3D coordinates of the UAV at time t, where H is the fixed flight altitude. Assuming the BS is located at fixed coordinates $w_G = [x_G, y_G, H_G]^T$, the distance $d(t)$ can be expressed as:

$$d(t) = \sqrt{(x(t) - x_G)^2 + (y(t) - y_G)^2 + (H - H_G)^2}\,. \quad (3)$$

Since the position of the UAV $w_U(t)$ changes continuously during task execution, the wireless transmission channel loss also varies with the change in distance between the UAV and the BS. At time slot t, the channel loss is expressed as

$$|h(t)|^2 = \beta_0 d(t)^{-\alpha}\,, \quad (4)$$

where β_0 is the fading coefficient, α is the path loss exponent, and $d(t)$ is the distance between the UAV and the BS.

If the UAV's transmission power is P_t, then the signal-to-noise ratio (SNR) at time t during transmission is given by

$$\rho(t) = \frac{P_t |h(t)|^2}{N_0 B}\,, \quad (5)$$

where B is the uplink transmission bandwidth, and N_0 is the power spectral density of the noise.

The UAV's uplink transmission rate can be expressed as

$$R_{up}(t) = B \log_2 \left(1 + \frac{P_t |h(t)|^2}{N_0 B}\right)\,. \quad (6)$$

The transmission delay ℓ_{tran} is related to the dimension of the semantic features K processed by the UAV and the uplink transmission rate $R_{up}(t)$

$$\ell_{tran} = \frac{K}{R_{up}(t)}\,. \quad (7)$$

For transmission energy consumption, the UAV processes the captured image and compresses the data into K semantic feature symbols. Therefore, the transmission energy at time t determined by the transmission power P_t and the uplink duration ℓ_{tran}, is given by

$$E_{tran} = \ell_{tran} P_t = \frac{K}{R_{up}(t)} P_t\,. \quad (8)$$

C. Task Evaluation Metric and Computation Model

For the image classification task considered in this work, the task execution accuracy is used as the evaluation metric, denoted as ξ. Since the accuracy of task execution depends on the choice of semantic model i, the number of transmitted semantic feature symbols K, and the SNR ρ, we evaluate task execution accuracy ξ by running the semantic models under different combinations of these factors. Specifically, we test various semantic models with different numbers of transmitted semantic feature symbols under varying SNRs on a dataset to obtain statistical results of task execution accuracy ξ. Based on the obtained accuracy values, the task execution accuracy can be modeled as a function of the number of transmitted semantic feature symbols and SNR, i.e., $\xi = f(i, K, \rho)$.

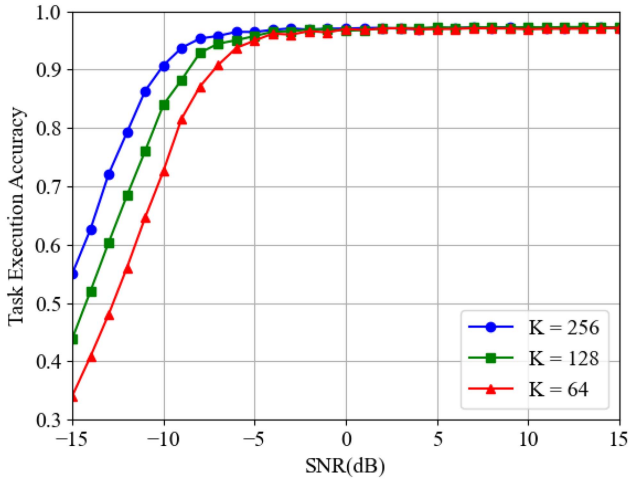


Fig. 3. Relationship between task execution accuracy, K, and SNR.

We obtain the relationship $f_i(K, \rho)$ between task execution accuracy ξ_i, number of semantic feature symbols K, and SNR by fixing semantic model and varying the number of transmitted semantic feature symbols and the SNR, as illustrated in Fig. 3. It can be observed from Fig. 3 that under high SNR conditions, different numbers of semantic feature symbols K have minimal impact on accuracy ξ. However, under low SNR conditions, a decrease in the number of transmitted semantic feature symbols K results in a more significant degradation in task execution accuracy ξ. Since K is a discrete variable, a curve-fitting method can be used to model the task execution accuracy ξ as a continuous function of SNR ρ for a given K.

The semantic model is based on the Transformer-based DeiT-small architecture. The baseline model uses a Transformer encoder with 12 layers [123], but in practical scenarios, the number

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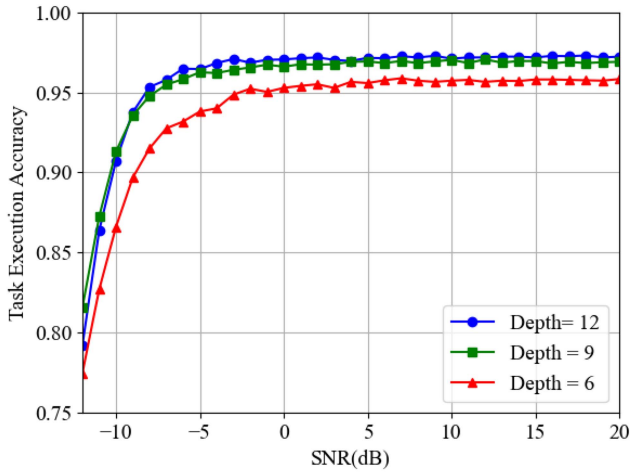


Fig. 4. Relationship between task execution accuracy, depth, and SNR.

of encoder layers can be adjusted up or down depending on resource constraints and task requirements. Since UAVs are energy-limited, reducing the number of encoder layers can decrease computational energy consumption at the cost of reduced accuracy. Therefore, UAVs may select an appropriate semantic model through trade-offs. In this study, semantic models with different Transformer encoder depths (6, 9, and 12) are evaluated using the same number of transmitted semantic feature symbols K under varying SNRs. The relationship $f_K(i,\rho)$ between model selection i, SNR, and task execution accuracy ξ is shown in Fig. 4. It can be seen that reducing the number of encoder layers leads to some accuracy ξ loss under high SNR conditions.

Each time the UAV makes a decision i, it selects one semantic model for local processing. Let f denote the CPU clock frequency of the UAV, and let c represent the number of floating-point operations (FLOPs) that can be executed per CPU cycle, which is determined by the hardware architecture. Let C_i denote the number of FLOPs required to process one sample using the i-th semantic model, which reflects the computational complexity of that model. Specifically, C_i is obtained by counting the floating-point operations of all layers involved in one forward inference pass of the selected semantic model. For fully connected layers, the FLOPs are calculated according to the input and output dimensions. For the Transformer-based semantic encoder, the FLOPs mainly come from multi-head self-attention and feed-forward network layers. Let D denote the number of samples processed in each time slot. Then, the total number of CPU cycles required for local inference in one time slot is given by DC_i/c.

Following the widely used dynamic power consumption model of CMOS processors, the computation energy consumption is proportional to the number of CPU cycles and the square of the CPU frequency [90] [124]. Therefore, the computational energy consumption of the UAV can be expressed as

$$E_{comp} = kDf^2 C_{i/}c\,, \tag{9}$$

where k is the effective switched capacitance coefficient related to the processor hardware.

Therefore, the total energy consumption of the UAV consists of transmission energy and computation energy, which can be expressed as

$$E = E_{tran} + E_{comp}\,. \tag{10}$$

Since this paper mainly focuses on the trade-off among communication, computation, and task performance in task-oriented semantic transmission, propulsion energy is not explicitly modeled. In the considered low-altitude inspection scenario, the UAV maintains a fixed flight altitude and follows relatively smooth trajectories without aggressive maneuvers. Under the same flight-altitude setting and similar speed constraints, the average propulsion energy consumed per unit travel distance can be regarded as relatively stable [125]. Therefore, the difference in propulsion energy can be approximately reflected by the difference in their total travel distances. In our work, the travel distances of the compared trajectories vary within a limited range. Based on this, propulsion energy can be regarded as a common background cost with limited variation, which does not significantly affect the comparison and performance evaluation of different optimization schemes [121].

Regarding computation latency ℓ_{comp}, since the BS is assumed to be equipped with high-performance servers, the processing delay at the BS is considered negligible compared to the UAV. Therefore, the computation delay mainly stems from the semantic inference on the UAV. The computation delay at the UAV can be expressed as

$$\ell_{comp} = DC_i/cf\,. \tag{11}$$

Accordingly, the total latency for performing an image transmission and executing the subsequent downstream task is given by

$$\ell = \ell_{tran} + \ell_{comp}\,. \tag{12}$$

D. Trajectory Planning

The UAV is required to sequentially pass through specific waypoints $\mathbf{w}_j (j = 1, 2, 3, 4\ldots)$ to complete image capture tasks. Therefore, the trajectory reward of the UAV is defined as

$$R = \begin{cases} -P + C & if\ \mathbf{w}_U = \mathbf{w}_j,\ j = 1, 2, 3, 4\ldots, \\ -P & else\,, \end{cases} \tag{13}$$

where $-P$ denotes the per-step penalty for the UAV, and C represents the reward received upon reaching a designated waypoint.

The UAV acceleration $\mathbf{a}(t)$ is selected as the basic control variable for trajectory optimization. Accordingly, the UAV velocity $\mathbf{v}_U(t)$ and position $\mathbf{w}_U(t)$ are updated as follows:

$$\mathbf{v}_U(t) = \mathbf{v}_U(t-1) + \mathbf{a}(t)\delta t\,, \tag{14}$$

$$\mathbf{w}_U(t) = \mathbf{w}_U(t-1) + \mathbf{v}_U(t)\delta t\,, \tag{15}$$

where δt denotes the duration of each time slot. To ensure operational safety and stability, the UAV's velocity is constrained within a predefined threshold range

$$0 \le \mathbf{v}_U(t) \le \mathbf{v}_{\max}\,. \tag{16}$$

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Specifically, The UAV trajectory determines its position $\mathbf{w}_U(t)$,which directly changes the distance $d(t)$ between the UAV and the base station. Based on the aforementioned large-scale channel model, this distance variation further affects the channel coefficient $h(t)$ and the SNR $\rho(t)$. Since the task execution accuracy is modeled as:

$$\begin{aligned}\xi(t) &= f(i(t), K(t), \rho(t)) \\ &= f\left(i(t), K(t), \frac{P_t \beta_0 d(t)^{-\alpha}}{N_0 B}\right). \end{aligned} \tag{17}$$

Therefore, the trajectory planner incorporates channel effects through the trajectory-dependent task accuracy, which is determined by the UAV-BS distance and the resulting SNR.

E. Optimization Problem Formulation

This paper aims to balance task execution performance and UAV energy consumption by jointly optimizing the UAV trajectory, allocating transmission power P, determining the number of transmitted semantic feature symbols K, and selecting the semantic model type i. Therefore, the optimization objective is formulated as

$$\max_{i,P,K,\mathbf{a}} \beta_1 \xi - \beta_2 E + R\,, \tag{18}$$

where β_1 and β_2, are the weighting coefficients for task execution accuracy and energy consumption, respectively.

The corresponding optimization problem is modeled as

$$\max_{i,P,K,\mathbf{a}} \beta_1 \xi - \beta_2 E + R, \tag{19}$$

$$\text{s.t.} \quad P_{\min} \le P(t) \le P_{\max}, \tag{19a}$$

$$\xi \ge \xi_{\min}, \tag{19b}$$

$$\ell \le \ell_{\max}, \tag{19c}$$

$$i \in I, \tag{19d}$$

$$K \in \mathcal{K}, \tag{19e}$$

$$0 \le \mathbf{v}_U(t) \le \mathbf{v}_{\max}, \tag{19f}$$

where constraint (19a) restricts the transmission power within the allowable range; (19b) and (19c) ensure the required accuracy and transmission delay constraints, with ξ_{min} denoting the minimum required task execution accuracy and ℓ_{max} representing the maximum allowable transmission delay. Constraint (19d) defines the set of selectable semantic model types, and (19e) specifies the allowable number of transmitted semantic feature symbols. (19f) constrains the UAV's velocity within a predefined threshold range.

IV. Trajectory Planning and Resource Allocation Algorithm

Considering that the task execution accuracy ξ depends on the UAV trajectory, semantic model selection, number of semantic feature symbols and transmission power, the optimization problem involves coupled variables. To manage the complexity of this joint optimization, we adopt an alternating optimization strategy. Specifically, we decompose the problem into two subproblems.

A. Trajectory Optimization

Under a fixed resource allocation policy P, K, i, we optimize the UAV trajectory to enhance the task execution accuracy and trajectory planning reward. Since the task execution accuracy ξ depends on the SNR determined by the UAV-BS distance, this trajectory optimization subproblem implicitly captures the mobility-induced large-scale channel variations during trajectory generation.

$$\max_{\mathbf{a}} \quad \beta_1 \xi + R, \tag{20}$$

$$\text{s.t.} \quad \xi \ge \xi_{\min}, \tag{20a}$$

$$\ell \le \ell_{\max}, \tag{20b}$$

$$0 \le \mathbf{v}_U(t) \le \mathbf{v}_{\max}. \tag{20c}$$

The problem is formulated as a MDP,

Action space: The UAV's acceleration $\mathbf{a}$ at each time slot generated using a Diffuser model.

State space: The UAV's current position and velocity.

Reward function: The UAV needs to satisfy waypoint constraints while enhancing the accuracy of semantic inference from images. Therefore, the reward function is defined as the objective function of the sub-problem.

In this trajectory planning problem, the UAV trajectory is subject to multiple waypoint $\mathbf{w}_j$ constraints, which makes the planning problem highly complex. Online RL requires repeated environment interactions, which may introduce high training overhead and unstable exploration. In contrast, offline RL learns from a fixed pre-collected dataset without additional online exploration, thereby reducing training overhead and improving learning stability. Therefore, we develop an offline RL algorithm based on the Diffuser model to optimize the UAV trajectory [50].

Compared with conventional RL methods, the Diffuser-based offline RL framework has the following advantages,

- The diffusion model can generate a complete trajectory sequence through conditional denoising, rather than making only one-step decisions.
- The offline RL algorithm learns from pre-collected trajectory data, avoiding unstable online exploration and reducing training overhead.
- The conditional diffusion framework has stronger generalization capability, enabling adaptation to different waypoint settings, start/end positions, and communication environments.

The offline trajectory dataset is first constructed by specifying the start point, destination, and waypoint constraints, and then generating trajectories with varying quality using heuristic planning strategies together with random perturbation factors. Therefore, the dataset contains both relatively good and relatively poor trajectories, which enables the diffusion model to learn the underlying distribution of feasible trajectory patterns under different task conditions.

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Diffusion probabilistic models formulate the data generation process as an iterative denoising process $p_\theta(\tau^{n-1} \mid \tau^n)$, where $n \sim \mathcal{U}\{1, 2, \ldots, N\}$ denotes the diffusion sampling time step and τ^n represents the noisy trajectory after n sampling time steps. This denoising is the reverse of a forward diffusion process that gradually corrupts the data by adding noise.

In the forward diffusion process, Gaussian noise is gradually added to the noiseless trajectory sample according to

$$q(\tau^n \mid \tau^{n-1}) = \mathcal{N}\left(\tau^n; \sqrt{1-\beta_n}\,\tau^{n-1}, \beta_n \mathbf{I}\right), \tag{21}$$

where $\beta_n \in (0, 1)$ is the variance schedule at diffusion step n. By recursively applying the above process, the clean trajectory is progressively corrupted into a noisy trajectory τ^N that approximately follows a Gaussian distribution.

In the reverse generation process, the model learns a parameterized denoising transition to recover the clean trajectory τ^0 from the noisy trajectory τ^N. The data distribution of the trajectory generated by the model is expressed as

$$p_\theta(\tau^0) = \int p(\tau^N) \prod_{n=1}^{N} p_\theta(\tau^{n-1} \mid \tau^n) \mathrm{d}\tau^{1:N}, \tag{22}$$

where $p(\tau^N)$ denotes the standard Gaussian prior distribution, τ^N denotes the noisy trajectory sample, and τ^0 represents the final generated trajectory obtained after the reverse denoising process.

The reverse process $p_\theta(\tau^{n-1} \mid \tau^n)$ is often parameterized as Gaussian distribution with covariances fixed and dependent on the timestep

$$p_\theta(\tau^{n-1} \mid \tau^n) = \mathcal{N}(\tau^{n-1} \mid \mu_\theta(\tau^n, n), \Sigma^n), \tag{23}$$

where Σ^n denotes the covariance of the reverse process. The denoising mean $\mu_\theta(\tau^n, n)$ can be calculated using a closed-form formula [126]

$$\mu_\theta(\tau^n, n) = \frac{1}{\sqrt{\beta_n}} \left(\tau^n - \frac{\alpha_n}{\sqrt{1-\bar{\alpha}_n}} \epsilon_\theta(\tau^n, n)\right), \tag{24}$$

where α_n, β_n and $\bar{\alpha}_n$ are predefined coefficients of the diffusion process related to the noise schedule. $\epsilon_\theta(\tau^n, n)$ is the noise gradient trained through the model.

During training, a simplified objective function is used

$$\mathcal{L}(\theta) = \mathbb{E}_{n, \epsilon, \tau^0}\left[\|\epsilon - \epsilon_\theta(\tau^n, n)\|^2\right], \tag{25}$$

where $\epsilon \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$ is a noise target.

By combining RL with the iterative denoising process of diffusion models, we enable sampling from a perturbed distribution

$$\tilde{p}_\theta(\tau) \propto p_\theta(\tau) h(\tau). \tag{26}$$

The function $h(\tau)$ serves as a perturbation or guidance function, which can incorporate desired outcomes, such as specified targets or optimization objectives.

Let O_t be a binary random variable indicating the optimality of the trajectory at time step t, with the probability defined as

$$p(O_t = 1) = \exp(r(s_t, a_t)), \tag{27}$$

Algorithm 1: Training and Guided Sampling for Diffusion-based Trajectory Generation.

Input: Diffusion model μ_θ, time steps N, guidance function ∇J, guidance scale ω

// Training phase

while not converged **do**
  Sample real trajectory τ^0 from dataset
  Sample time step $n \in [1, N]$, noise $\epsilon \sim \mathcal{N}(0, I)$
  Generate noisy trajectory after n sampling time steps $\tau^n \leftarrow \sqrt{\bar{\alpha}_n}\tau^0 + \sqrt{1-\bar{\alpha}_n} \cdot \epsilon$
  Using diffusion model to predict noisy $\epsilon_\theta(\tau^n, n)$
  Compute loss L between the predicted noise ϵ_θ and the real ϵ noise using (25)
  Update θ via gradient descent on L
end while

//Planning phase

Initialize $\tau^N \sim \mathcal{N}(0, I)$
for $n = N$ to 1 **do**
  Predict noise $\epsilon_\theta(\tau^n, n)$
  Calculate the denoising mean $\mu_\theta(\tau^n, n)$ through (24)
  Iteratively denoise according to:

$$\tau^{n-1} \sim \mathcal{N}\left(\mu_\theta(\tau, n) + \omega \Sigma^n \nabla J(\mu_\theta), \Sigma^n\right)$$

end for
return Generated trajectory τ_0

where $r(s_t, a_t)$ denotes the reward at time step t.

By setting the perturbation function as

$$h(\tau) = p(O_{1:T} \mid \tau), \tag{28}$$

we sample trajectories conditioned on optimality, thereby reformulating the RL problem as a conditional trajectory sampling problem.

When $p(O_{1:T} \mid \tau_n)$ is sufficiently smooth, the reverse diffusion transition can be approximated by a Gaussian distribution [127]

$$p_\theta(\tau^{n-1} \mid \tau^n, O_{1:T}) \approx \mathcal{N}(\tau^{n-1}; \mu + \Sigma g, \Sigma), \tag{29}$$

where μ and Σ are the parameters of the original reverse process transition $p_\theta(\tau^{n-1} \mid \tau^n)$, and

$$\begin{aligned} \mathrm{g} &= \nabla_\tau \log p(\mathcal{O}_{1:T} \mid \tau)|_{\tau=\mu} \\ &= \sum_{t=0}^{T} \nabla_{s_t, a_t} r(s_t, a_t)|_{(s_t, a_t)=\mu_t} = \nabla \mathcal{J}(\mu). \end{aligned} \tag{30}$$

This formulation enables diffusion-based policies to sample trajectories that are not only likely under the learned model but are also optimized with respect to the RL objective, naturally supporting flexible goal or reward conditioning.

B. Resource Allocation Optimization

The second subproblem focuses on optimizing the resource allocation policy to better adapt to local channel conditions and system constraints under the condition of UAV trajectories generated by the Diffuser model. After the trajectory is

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generated, the resource allocation module further adapts the communication and computation resources according to the current trajectory-dependent channel state, thereby improving the robustness of task execution under varying and low-SNR channel conditions. The problem is formulated as

$$\max_{i,P,K} \quad \beta_1 \xi - \beta_2 E, \tag{31}$$

$$\text{s.t.} \quad P_{\min} \le P(t) \le P_{\max}, \tag{31a}$$

$$\xi \ge \xi_{\min}, \tag{31b}$$

$$\ell \le \ell_{\max}, \tag{31c}$$

$$i \in I, \tag{31d}$$

$$K \in \mathcal{K}. \tag{31e}$$

To solve the joint optimization of computational and communication resources, we employ the PPO algorithm. The problem is modeled as a MDP with the following components.

Action Space: The UAV's action space consists of discrete variables including the choice of semantic model $i_t \in I$ and the number of semantic feature symbols K_t to transmit, as well as a continuous variable the transmission power P_t. The joint action space can thus be expressed as $a_t^{RA} = (i_t, K_t, P_t)$.

State Space: The UAV makes decisions based on the transmission channel gain, so the state at each time step is defined as $s_t^{RA} = (K_{t-1}, R_{t-1}, h_t)$, where K_{t-1} is the number of semantic feature symbols transmitted in the previous time slot, R_{t-1} is the previous uplink transmission rate, and h_t is the current channel gain, which dynamically varies with the UAV's distance to the BS.

Reward Function: The reward is designed as the weighted sum of the accuracy of task execution and energy consumption, which is the objective of the subproblem.

To satisfy the accuracy constraint (31b) and delay constraints (31c), a penalty mechanism is introduced. If the transmission delay exceeds the threshold or the task execution accuracy falls below the minimum acceptable level, a penalty value η is assigned to the reward. The modified reward function is given by

$$r_t^{RA} = \begin{cases} \eta & \text{if } \xi_t < \xi_{\min} \text{ or } \ell_t > \ell_{\max}, \\ \beta_1 \xi_t - \beta_2 E_t & \text{otherwise.} \end{cases} \tag{32}$$

This paper adopts the PPO-based algorithm to solve the joint computation and communication resource allocation problem. The policy network $\pi_\theta(a_t^{RA}|s_t^{RA})$ maps the environmental state s_t^{RA} at time step t to the action space a_t^{RA}, where θ represents the parameters of the neural network. The objective of $\pi_\theta(a_t^{RA}|s_t^{RA})$ is to generate a Gaussian distribution characterized by a certain mean and variance, from which actions $a_t^{RA} = (i_t, K_t, P_t)$ are sampled to make decisions.

A value network $Q_\pi(s_t^{RA}, a_t^{RA})$ is introduced to evaluate the expected value of the current reward and subsequent rewards obtained by executing action a_t^{RA} in state s_t^{RA}. The value function $V_\psi(s_t^{RA})$ for a given state s_t^{RA} is defined as

$$V_\psi(s_t^{RA}) = \sum_a \pi_\theta(a_t^{RA}|s_t^{RA}) \cdot Q_\pi(s_t^{RA}, a_t^{RA}). \tag{33}$$

Apply the action a_t^{RA} to the environment and calculate the reward value. After executing the action a_t^{RA} in the current environment, update the state s_{t+1}^{RA} for the next time slot, and store the action, state, reward value of the current time slot and the state of the next time slot in the buffer $(a_t^{RA}, s_t^{RA}, r_t^{RA}, s_{t+1}^{RA})$.

After collecting $(a_t^{RA}, s_t^{RA}, r_t^{RA}, s_{t+1}^{RA})$ over T time steps, the Temporal Difference (TD) error and advantage function can be computed as

$$\delta_t = r_t^{RA} + \gamma V_\psi(s_{t+1}^{RA}) - V_\psi(s_t^{RA}), \tag{34}$$

$$\hat{A}_t = \delta_t + (\gamma\lambda)\delta_{t+1} + \cdots + \cdots + (\gamma\lambda)^{T-t+1}\delta_{T-1}, \tag{35}$$

where γ is the discount factor, which represents the degree of influence of future rewards on current decision. λ is the smoothing parameter.

The training objective for the value network $V_\psi(s_t^{RA})$ is to minimize the difference between the predicted value and the actual discounted return. The corresponding loss function is

$$L_t^{VF} = (V_\psi(s_t^{RA}) - V_t^{\text{targ}})^2, \tag{36}$$

where $V_t^{\text{targ}} = r_t^{RA} + \gamma V_\psi(s_{t+1}^{RA})$ denotes the estimated discounted return.

To constrain the policy update step, PPO introduces a clipped objective function to improve the robustness and stability [128].

Algorithm 2: PPO-based Joint Computation and Communication Resource Allocation.

Input: number of steps per episode N_STEP, clipping threshold ϵ

Initialize policy network π_θ and value network $V_\psi(s_t^{RA})$
for episode in $MAX_EPISODE$ **do**
  Initialize environment state s_0^{RA}
  for step $= 1$ to N_STEP **do**
    Observe current state s_t^{RA}
    Input s_t^{RA} into policy network π_θ, obtain mean μ and variance σ
    Sample action $a_t^{RA} \sim \mathcal{N}(\mu, \sigma^2)$
    Apply action a_t^{RA} to environment, compute reward r_t^{RA} using objective function (32)
    Receive next state s_{t+1}^{RA} and reward r_t^{RA}
    Store $(s_t^{RA}, a_t^{RA}, r_t^{RA}, s_{t+1}^{RA})$ into replay buffer
    $s_t^{RA} \leftarrow s_{t+1}^{RA}$
  end for
  Select a batch of samples from the buffer
  Compute advantage estimates using (35)
  Update policy and value networks using loss in (38)
end for
return Trained policy network

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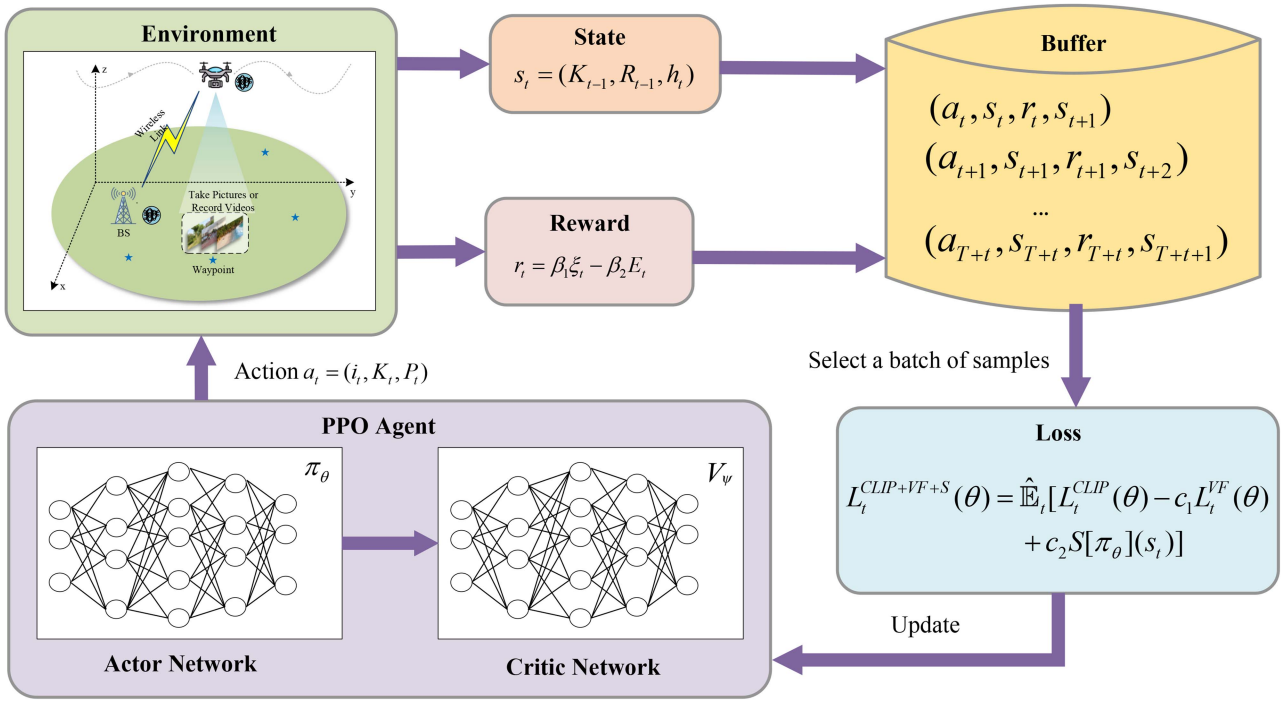


Fig. 5. The training process of PPO.

The modified objective function is

$$L^{CLIP}(\theta) = \hat{\mathbb{E}}_t[\min(r_t^{RA}(\theta)\hat{A}_t, clip(r_t^{RA}(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t)]. \tag{37}$$

By combining the clipped objective, the value loss, and an entropy bonus to encourage exploration, the final loss function becomes

$$L_t^{CLIP+VF+S}(\theta) = \hat{\mathbb{E}}_t\left[L_t^{CLIP}(\theta) - c_1 L_t^{VF}(\theta) + c_2 S[\pi_\theta](s_t^{RA})\right], \tag{38}$$

where c_1 and c_2 are weighting coefficients for the value loss and entropy bonus, respectively. $S[\pi_\theta]$ is the entropy bonus.

The training process are illustrated in Fig 5, and Algorithm 2 summarizes the overall PPO-based optimization procedure for jointly allocating computational and communication resources, aiming to maximize the weighted sum of task execution accuracy and energy efficiency.

V. Simulation Experiments

A. Experiment Setting

This paper considers an image transmission scenario in a remote area involving a UAV and a ground BS. The position of the BS is set at (3 km, 4 km, 0), while the UAV operates at a relative altitude of 0.1 km. The UAV's initial and final positions are configured as (0 km, 0 km, 0.1 km) and (3 km, 7 km, 0.1 km), respectively. Four waypoints are set along the UAV's trajectory. Fig. 6 shows the top view of the UAV's starting point, destination point, intermediate waypoints, and the BS.

The uplink bandwidth between the UAV and the BS is set to 1 MHz. The UAV's minimum and maximum transmission power P are 0.05 W and 0.5 W, respectively [129]. The number of transmitted semantic features is denoted as $K \in \{64, 128, 256\}$.

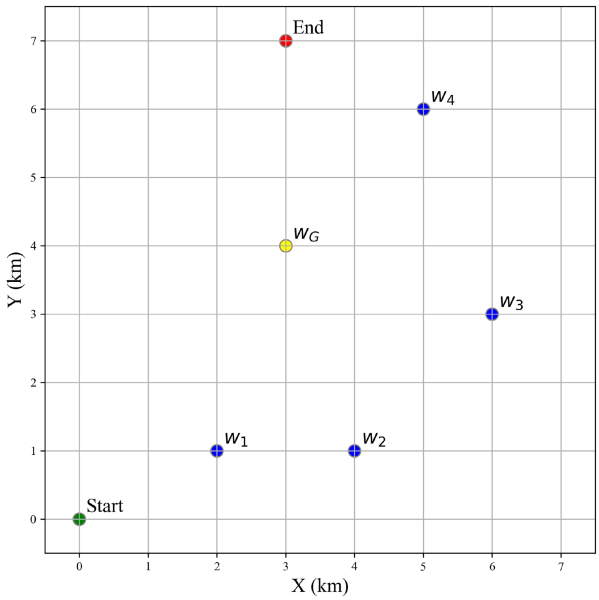


Fig. 6. The simulation points of UAV (top view).

The UAV executes tasks over discrete time slots, with each slot lasting 1 s. The maximum velocity $\mathbf{v}_{\max}$ is set to $30\,\mathrm{m/s}$. The required task execution accuracy threshold is 60%, and the maximum transmission delay is set to 0.25 seconds. The weights for the task execution accuracy β_1 and energy consumption β_2 in the objective function are both set to 0.5. The channel parameters include the noise power spectral density $N_0 = -169$ dBm/Hz, reference path loss $\beta_0 = -60$ dB, and path loss exponent $\alpha = 2.3$. The UAV computation parameters include the number of floating-point operations per clock cycle $c = 32$, the CPU frequency $f = 1.5$ GHz, and the CPU power

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TABLE I
REPRESENTATIVE STUDIES ON RESOURCE ALLOCATION AND TRAJECTORY OPTIMIZATION FOR UAV COMMUNICATIONS

Category	Ref.	Scenario	Optimization Objective	Optimized Variables	Methodology
Traditional UAV MEC Optimization	[74]	UAV-assisted MEC	Energy consumption minimization	Resource partitioning, UAV trajectory and bit allocation	SCA
	[75]	UAV-assisted MEC	Energy consumption minimization	UAV Trajectory, the assignment coefficient of task and CPU's computational speed	CCCP
	[78]	UAV-mounted cloudlet for MEC	Energy consumption minimization	Bit allocation and cloudlet's trajectory	SCA
	[81]	Detecting and tracking task of UAVs in the maritime environment	Task latency minimization	Transmission power, local CPU frequency, offloading rate, and bandwidth allocation	The convex optimization and the genetic algorithm
	[87]	Dual UAV-assisted MEC systems	secure computing capacity maximization	Communication resources, computation resources, and UAVs' trajectories	BCD and a PBCD based algorithm
	[88]	UAV-assisted MEC	UAV energy consumption minimization	UAV trajectory, the user transmit power, and computation load allocation	Dinkelbach algorithm and SCA algorithm
RL-based UAV Offloading and Control	[89]	UAV-assisted edge computing	Energy consumption and total processing time minimization	UAV and task cluster matching and MEC offloading server optimization	K-Means and Q-learning
	[90]	UAV-assisted MEC systems	Energy consumption and delay minimization	Computing offloading strategy and the UAV trajectory	DDPG and PSO
	[91]	Multi-UAV cooperative MEC	Energy consumption minimization	Task offloading ratios and UAV trajectories	IPPO
	[92]	UAV-assisted MEC networks	Time-averaged secrecy capacity	UAV trajectory, task offloading decisions, and time allocation ratios	DQN
	[93]	UAV-assisted MEC systems	Energy consumption and time delay	Computational cost, 3D UAV placement and resource allocation	DDPG
	[94]	Multi-UAV enabled MEC networks	Task processing delay	Discrete computation modes, continuous 3D trajectories, and resource assignment	MADDPG, Coordinate Descent (CD) and Karush-Kuhn-Tucker (KKT) conditions
Semantic-aware UAV Communication	[11]	UAV semantic communication for image classification	The tradeoff between transmission latency and classification accuracy	Extraction of semantic blocks	DRL-based algorithm
	[47]	UAV image-sensing task-oriented scenario	The scores for transmission time and the received information amount	Transmit power, channel assignment, and the number of transmitted semantic symbols	HDCD-SC
	[95]	UAV semantic communication for image retrieval	Multi-user joint utility maximization	Semantic triplet weight and transmit power	Game theory-based algorithm
	[48]	UAV-enabled spectrum sharing semantic communication networks with OFDM	Semantic spectrum efficiency (S-SE) maximization	Subchannel assignment, semantic symbols assignment, transmit beamforming and trajectory of UAV	DDQN, DDPG
	[96]	UAV semantic communication for digital twin	Synchronization latency minimization	Synchronization indicator, semantic extraction factor, computation resource and transmission power	DRL-based algorithm
	[97]	Multi-UAV image semantic transmission	The transmission delay minimization	UAV trajectory, channel allocation, and semantic information selection	VD-MADRL
Our Work	—	UAV task-oriented semantic communication for image classification	Energy-task accuracy tradeoff	UAV Trajectory, the number of semantic feature symbols, semantic model, transmission power	Diffuser-based trajectory and PPO-based resource allocation

consumption coefficient $k = 10^{-28}$ [130]. The simulation parameters are summarized in Table III.

The semantic model is trained using the Aerial Image Dataset (AID) [131], which is designed for UAV-based scene classification. The dataset contains 10,000 aerial images across 30 different scene categories, such as airports, bare land, baseball fields, beaches, bridges, churches, and more.

B. Results and Analysis

Two conventional transmission schemes are set as baselines for comparison to evaluate the performance advantages of SemCom in UAV systems for image classification tasks. The baseline schemes are defined as follows,

Local Inference scheme (LI): The UAV performs the image classification task locally and only the classification results are transmitted to the BS through the channel. The final classification label needs to be transmitted over the wireless channel with finite coding redundancy, rather than being assumed to be received without error.

Raw Image Offloading scheme (RIO): The UAV does not perform local image processing, and the image data is transmitted directly to the BS, where the image classification task is fully executed.

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TABLE II
SUMMARY OF KEY NOTATIONS

Notation	Description
UAV Semantic Communication System	
$\mathbf{s}$	Original images information
$\mathbf{x}$	Transmitted semantic feature symbols at UAV
$\mathbf{y}$	Received semantic feature symbols at BS
$\mathbf{z}$	Extracted semantic information at UAV
$\hat{\mathbf{z}}$	recovered semantic information at BS
P	The final classification output
ξ	Image classification accuracy
i	The index of semantic models with different depth
K	The number of transmitted semantic feature symbols
H	The flight altitude of the UAV
$h(t)$	Channel gain between UAV and BS
$d(t)$	UAV-BS distance at time slot t
B	Available communication bandwidth
P_t	UAV transmit power at time slot t
$\rho(t)$	Signal-to-noise ratio (SNR) at time slot t
$R_{up}(t)$	UAV uplink transmission rate at time slot t
ℓ_{tran}	Transmission delay
E_{tran}	Transmission energy consumption
Computation and Energy Model	
f	CPU frequency of UAV
c	FLOPs CPU can perform per clock cycle
C_i	FLOPs of the i-th semantic model
E_{comp}	Computation energy consumption
E	Total UAV energy cost
ℓ_{comp}	Computation delay
ℓ	Total delay
Trajectory Planning	
w_U	Position of UAV
w_G	Position of BS
$\mathbf{v}_U$	Velocity of UAV
$\mathbf{a}$	Acceleration of UAV
T	Number of time steps
Reinforcement Learning	
s_t	State at time slot t
a_t	Action at time slot t
r_t	Reward at time slot t
π_θ	Policy network parameterized by θ
$V_\psi(s_t)$	Value function parameterized by ψ
γ	Discount factor
A_t	Advantage estimator at slot t

TABLE III
SIMULATION PARAMETER SETTINGS

Parameter	Value
N_0	$-169\,\mathrm{dBm/Hz}$
β_0	$-60\,\mathrm{dB}$
α	2.3
B	$1\,\mathrm{MHz}$
P	$0.05\,\mathrm{W} \sim 0.5\,\mathrm{W}$
c	32
f	$1.5\,\mathrm{GHz}$
k	10^{-28}
H	$100m$
$\mathbf{v}_{\max}$	$30\,\mathrm{m/s}$

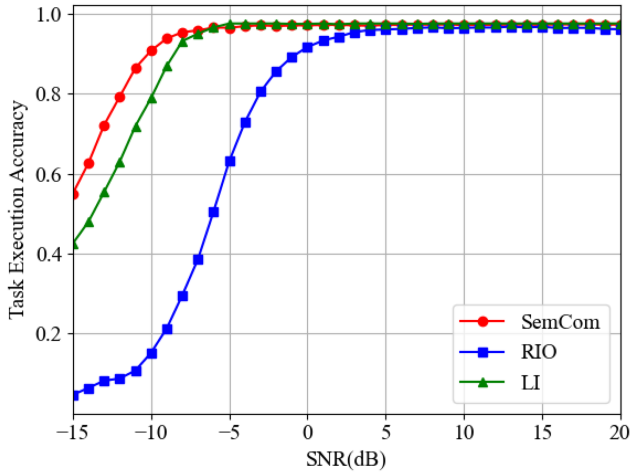
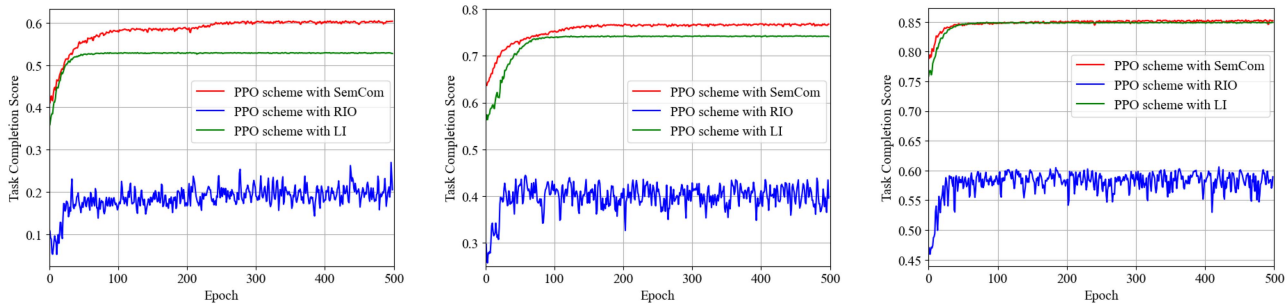


Fig. 7. Task execution accuracy of semantic and conventional communication schemes.

Fig. 7 shows the task execution accuracy of the semantic and conventional communication schemes varies with the SNR level. Under low SNR conditions, the SemCom scheme exhibits a significant performance improvement over conventional communication schemes. In particular, when the SNR is at -10 dB, the proposed semantic communication scheme improves task execution accuracy by 15% over the local inference scheme and achieves a 5-fold gain over the raw image offloading scheme. This demonstrates that, under extremely noisy channel conditions, raw image transmission is highly vulnerable to channel distortion. For the LI scheme, the 30-class classification result is encoded into 5 bits and transmitted with an adaptive repetition factor from 2 to 16 according to the channel condition. Although a repetition factor of 16 already introduces a relatively high redundancy level in practical low-latency short-packet transmission, the final label is still delivered through finite-redundancy hard-decision digital transmission. Under very low-SNR conditions, label bit errors may directly lead to an incorrect task result, resulting in a cliff effect. In contrast, JSCC-based SemCom transmits continuous semantic representations and usually exhibits more graceful degradation under noisy channels. This implies that SemCom can maintain high task execution accuracy under low SNR conditions, and that even in poor channel environments, it can achieve satisfactory performance without requiring a substantial increase in transmission power.

Similar to the SemCom scheme, three models with different levels of complexity are configured in the local inference scheme, each corresponding to distinct computational energy

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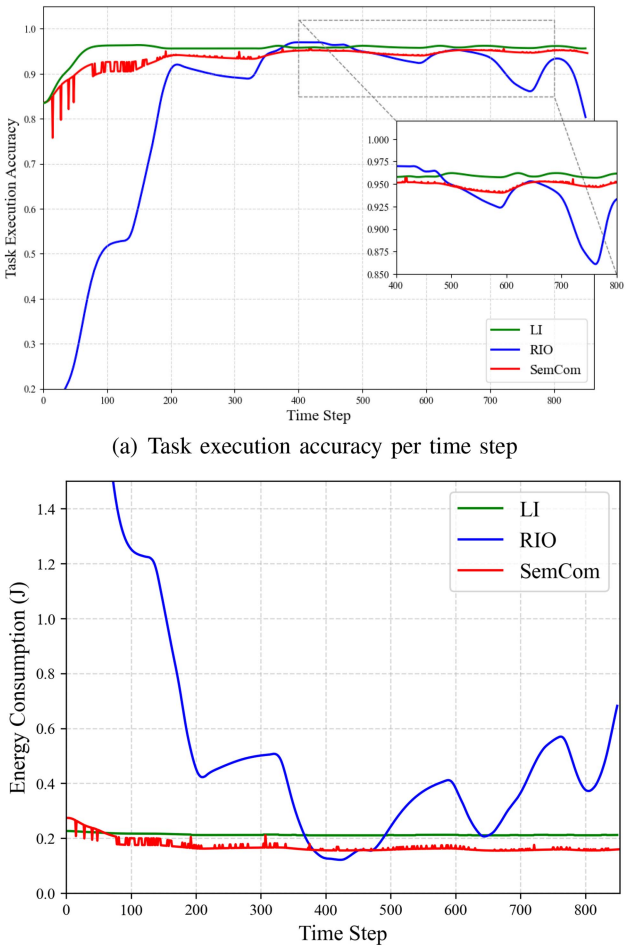
(a) $\beta_1 : \beta_2 = 0.5$ (b) $\beta_1 : \beta_2 = 1$ (c) $\beta_1 : \beta_2 = 2$

Fig. 8. Reward values over training epochs for different schemes.

consumption and task execution accuracy curves. In the local inference scheme, the computational energy consumption at the UAV side is neglected. These two conventional schemes are modeled by formulating an optimization problem, where the objective is defined as the weighted sum of task execution accuracy and energy consumption. The optimization problems under the local inference, raw image offloading, and SemCom schemes are solved using the PPO algorithm.

Fig. 8 presents the average reward per step of the three communication schemes varies with the training epochs. It is shown that the semantic communication scheme achieves the highest task completion score when the ratio of β_1 and β_2 is set to 0.5 or 1. As the ratio between β_1 and β_2 increases, the task completion performance of the local inference scheme gradually approaches that of the semantic communication scheme. In contrast, the raw image offloading scheme consistently exhibits the lowest task completion score across all tested settings. This is because, under the SNR considered in this experimental setting, the local inference and semantic communication schemes achieve comparable task execution accuracy. However, the local inference scheme incurs higher computational energy consumption. Therefore, as the weighting of energy consumption is reduced, the task performance scores of the local inference and semantic communication schemes become increasingly similar. The raw image offloading scheme requires substantially higher communication energy consumption and suffers from lower task execution accuracy, resulting in consistently inferior overall task scores under low SNR conditions.

Fig. 9 illustrates the UAV's task execution accuracy and energy consumption vary with the mission time. As shown in Fig. 9(a), the wireless channel experiences continuous dynamic variations as the UAV moves. The task execution accuracy of the raw image offloading scheme fluctuates significantly with the channel variations, whereas both the local inference scheme and the SemCom scheme demonstrates much higher stability, with local inference being slightly superior. Fig. 9(b) shows that the energy consumption of the local inference scheme is noticeably higher than that of the SemCom scheme, primarily due to its greater computational energy expenditure during local processing. In comparison, the raw image offloading scheme incurs substantially higher overall energy consumption, as transmitting large volumes of raw image data requires significant communication energy. Moreover, its energy consumption



(a) Task execution accuracy per time step

(b) Energy consumption per time step

Fig. 9. Task execution accuracy and energy consumption per time step.

exhibits greater fluctuations under varying channel conditions, reflecting the high sensitivity of wireless transmission cost to channel quality.

To evaluate the performance of the PPO algorithm in addressing the task-oriented resource allocation problem of UAV semantic communication systems, PPO is compared with various DRL baseline algorithms. Fig. 10 illustrates the average reward per step of the PPO, SAC [132], and TD3 [133] algorithms varies with the training epochs. It can be observed that the performance

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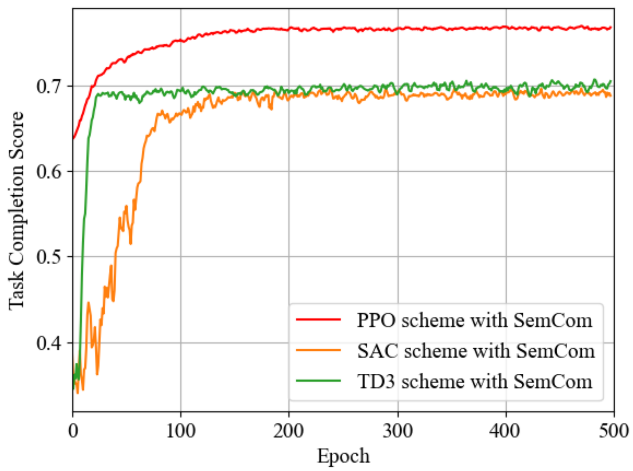


Fig. 10. Task completion score over training epochs for different RL algorithms.

TABLE IV
ALGORITHM COMPARISON TABLE

Algorithm	Diffuser	IQL	PPO
Reward	59.6729	53.6596	69.9044
Steps	643	461	468
Curvature Variance	0.4491	3.6819	4.8842
Acceleration Variance	2.61×10^{-7}	1.01×10^{-6}	6.31×10^{-7}

of PPO after convergence is significantly superior to that of the other two algorithms. By constraining the magnitude of policy updates, PPO algorithm can prevent large updates that could cause severe performance fluctuations, thereby enhancing the stability of policy optimization.

Fig. 11 illustrates the trajectories generated by the Diffuser algorithm, the online reinforcement learning algorithm PPO, and offline reinforcement learning algorithm IQL [134]. Table IV presents the trajectory rewards, total step lengths, and trajectory smoothness metrics obtained by the three algorithms. In terms of the reward function, PPO serves as a strong baseline because it can usually obtain a near-optimal policy after sufficient training in a specific environment. Nevertheless, PPO is less suitable for holistic multi-waypoint trajectory generation, since it tends to solve the trajectory planning problem in a segmented manner and then concatenate the generated trajectory segments. In contrast, Diffuser directly generates the entire trajectory through a unified conditional generation process. As reported in Table IV, the reward of Diffuser is close to that of PPO, indicating that Diffuser achieves near-optimal trajectory quality. Although Diffuser produces more trajectory steps and thus incurs higher time penalties, it still maintains a competitive reward, which demonstrates its effective global planning capability.

For practical UAV deployment, smoother trajectories imply more stable flight attitudes, lower maneuvering energy consumption, improved dynamic feasibility, and reduced mechanical wear on the UAV platform. To quantitatively evaluate trajectory smoothness, we further introduce the variance of curvature and the variance of acceleration as smoothness metrics. As reported in Table IV, Diffuser achieves the lowest curvature variance and acceleration variance, significantly outperforming IQL and PPO. The reason is that Diffuser generates the entire trajectory in a holistic and continuous manner, rather than optimizing local waypoint intervals separately. In contrast, PPO can only generate trajectories in a segmented manner for individual waypoint intervals, which may lead to noticeable bending at waypoint transitions. Although IQL is also an offline RL method, its value-based policy learning mechanism does not explicitly model global trajectory continuity, making it more prone to local irregularities and suboptimal turning behaviors.

Fig 11(d) illustrates the trajectory generated by Diffuser in an environment outside the dataset (with different waypoints set). It can be seen that the Diffuser-based offline RL algorithm is capable of generating suitable trajectories in unseen environments without retraining the model. Although the reward value of Diffuser-generated trajectory is slightly lower than that of PPO, Diffuser possesses the generalization ability to adapt to various waypoint environments a capability that online reinforcement learning algorithms lack.

C. Near-Optimality Verification in Reduced-Scale Scenario

To further evaluate the near-optimality of the proposed Diffuser-based trajectory planner, an additional reduced-scale experiment is conducted in a simplified environment where exact optimization is computationally tractable. Specifically, a representative trajectory segment from the original large-scale scenario is selected, with the UAV starting point set to (2,1) and destination point set to (4,1), while the BS location remains unchanged. In this reduced-scale setting, both the state space and action space are discretized, and the reward function is a weighted sum of task completion score and destination-distance reward

$$\mathcal{R} = \sigma_1 \xi + \sigma_2 \phi(d_g) + R, \tag{39}$$

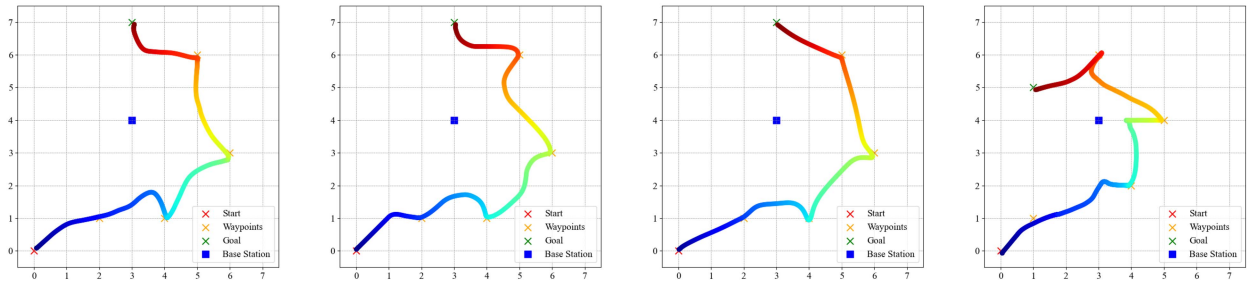
where σ_1 and σ_2 are weighting coefficients, set to 0.5 and 1, respectively, to balance the contributions of task quality and distance-based reward. R represents the trajectory reward defined in (13), where P is set to 0.25, C is set to 100. d_g represents the distance to the goal and the function $\phi(d_g)$ is a distance-based reward term that encourages the UAV to approach the destination by assigning higher rewards to shorter distances. Specifically, it is defined as

$$\phi(d_g) = \exp\left(-d_g\right), \tag{40}$$

Based on this discretized environment, dynamic programming (DP) is employed to compute the optimal trajectory as a benchmark solution. For fair comparison, Diffuser, IQL, and PPO are also evaluated in the same reduced-scale environment.

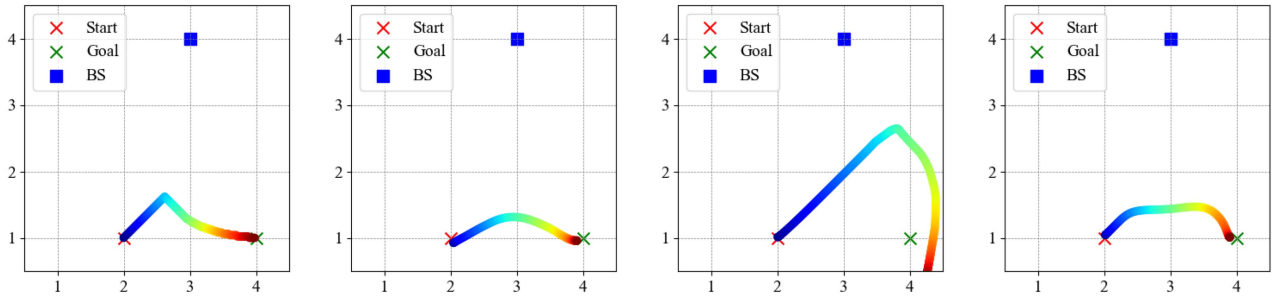
Fig. 12 illustrates the trajectories generated by dynamic programming, Diffuser, IQL, and PPO in the reduced-scale scenario, and Table V summarizes the trajectory rewards of different algorithms. As expected, dynamic programming attains the highest reward, serving as the optimal benchmark under the discretized setting. However, it is only applicable to discretized

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(a) Trajectory generated by diffuser (b) Trajectory generated by IQL (c) Trajectory generated by PPO (d) Trajectory generated by diffuser in a new environment

Fig. 11. Trajectories generated by Diffuser, IQL, and PPO.



(a) Trajectory generated by DP (b) Trajectory generated by diffuser (c) Trajectory generated by IQL (d) Trajectory generated by PPO

Fig. 12. Trajectories generated by DP, Diffuser, IQL, and PPO in reduced-scale scenario.

TABLE V
ALGORITHM COMPARISON TABLE

Algorithm	Diffuser	IQL	PPO	DP
Reward	136.0381	53.6596	140.1603	144.5660
Steps	98	200*	92	99

* Indicates the algorithm failed to reach the goal within the maximum limit (200 steps).

environments and is therefore limited to small-scale scenarios, while its computational complexity becomes prohibitively high in large-scale continuous settings. Both Diffuser and PPO achieve performance close to the DP solution, indicating their ability to learn near-optimal trajectory planning strategies in this simplified environment. However, PPO is an online reinforcement learning method that typically learns policies tailored to a specific task setting and generates single-goal-oriented trajectories, which limits its adaptability to more complex multi-waypoint scenarios and unseen environments. Meanwhile, IQL shows the lowest performance, as it lacks global planning capability and is more prone to being trapped in local optima. As a result, IQL fails to reach the destination within the maximum step constraint (200 steps), leading to the lowest trajectory reward among all methods.

The results demonstrate that the proposed Diffuser-based planner exhibits strong global planning capability, so as to be well-suited for large-scale continuous environments.

VI. CONCLUSION

In this paper, we have investigated task-oriented semantic communication for UAV image transmission in edge scenarios. To reduce UAV's energy consumption, we have proposed a UAV-adaptive task-oriented semantic communication framework (UA-TSC) with a semantic model lightweighting strategy. We have further developed a Diffuser-based trajectory optimization scheme and a PPO-based computation and communication resource allocation scheme to realize the trade-off between task execution accuracy and energy consumption. The simulation results demonstrate that the proposed UA-TSC scheme improves task execution accuracy by 15% over the local inference scheme and achieves a 5-fold improvement over the raw image offloading scheme at an SNR of -10 dB. In addition, the PPO-based resource allocation strategy consistently achieves higher task completion score than SAC and TD3 algorithm. The Diffuser-based trajectory optimizer exhibits stronger smoothness and generalization capability compared with IQL and PPO algorithm. In future work, we will extend our framework to semantic-based intelligent connectivity in low-altitude UAV networks.

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